

The transshipment problem

The maximum flow problem

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Optimization: principles and algorithms

The maximum flow problem

The maximum flow problem was motivated by the analysis performed by the American army of the railway network operated by the Soviet Union across Eastern Europe during the Cold War.

We have a network, where each arc has a capacity

Consider an origin o and a destination d

What is the maximum amount of flow that the network can carry from o to d .

The maximum flow problem

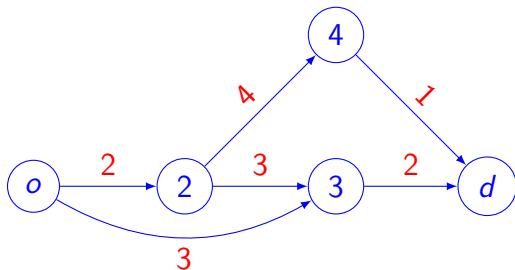
Show the 3 possible paths

$o \rightarrow 2 \rightarrow 4 \rightarrow d$: 1 unit of flow

$o \rightarrow 3 \rightarrow d$: 2 units of flow

$o \rightarrow 2 \rightarrow 3 \rightarrow d$: saturated.

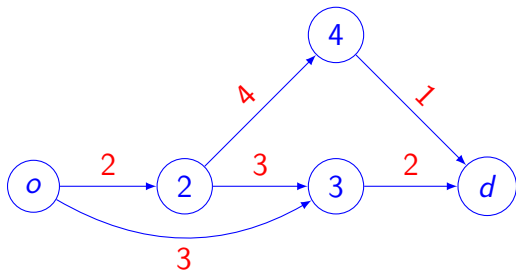
The maximum flow problem



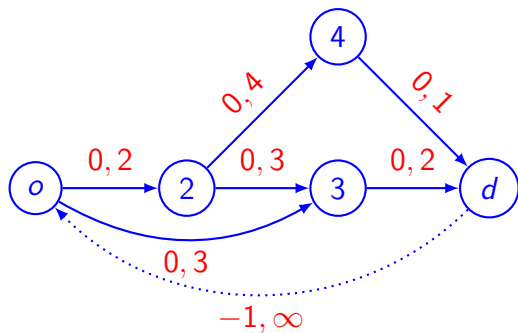
Transshipment

*Explain the idea of the artificial arc with infinite capacity
Add the cost on each arc*

Transshipment



Transshipment



- ▶ cost on the artificial arc: -1
- ▶ cost on every other arc: 0
- ▶ supply/demand for each node: 0
- ▶ lower bound on each arc: 0
- ▶ upper bound on the artificial arc: ∞
- ▶ upper bound on every other arc: *from the problem definition*

Transshipment

$$\min_{x \in \mathbb{R}^{n+1}} -x_{do}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{AU}(d,o)} x_{ij} - \sum_{k|(k,i) \in \mathcal{AU}(d,o)} x_{ki} = 0, \quad \forall i \in \mathcal{N}, \quad (1)$$

$$x_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A}, \quad (2)$$

$$x_{ij} \geq 0, \quad \forall (i,j) \in \mathcal{A}, \quad (3)$$

$$x_{do} \geq 0, \quad (4)$$

Transshipment

Summary

The maximum flow problem can be written as a transshipment problem

An artificial arc, carrying every flow that reaches the destination back to the origin, plays the role of a counter

The associated vector of flow must therefore be a circulation

Again, the max flow problem share the properties of the transshipment problem.